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SAMPLE CMT LEVEL I EXAM

PREFACE:

The following document is a sample exam only, not an actual exam of current or former designation in any way. The questions have been constructed/ phrased in a manner that is similar to past exams, but none of these questions, to the knowledge of the MTA staff, have ever appeared on an exam. The objective of this sample exam is to give as clear an indication as possible of the question structure and language one can expect in the exam. Every reasonable effort has been taken to make these exam questions representative of what candidates will see in the exams, however the following caveats should be noted.

- Exam questions change constantly. There may be up to 25% variation in the question mix in the Level I exams from one test cycle to the next. These sample questions are patterned after a limited number of questions drawn from a much larger pool of actual exam questions. This further increases the chances that the questions appearing in the exam may be significantly different from those contained in this sample. Candidates are cautioned **not to expect** the questions from the previous years' exams to appear in the current year.
- Candidate knowledge and experience will vary. The actual exam questions may be more difficult than those represented here, though that is unintentional. A question may be more difficult for one candidate and not another.
- The sample questions cover a variety of topics; however, in the actual exam the weighting may vary, i.e. the proportion of questions drawn from different topics may be different from the pattern followed in this sample paper.
- The actual exam consists of 135 questions of which 120 are scored items. Candidates are allotted two and half hours to take the Level I exam. This sample exam should be completed in 2.5 hours to better simulate performance on the exam.

Please note that this sample question booklet has been prepared completely independently from the exam to ensure that the questions that will appear in the actual exam paper remain secure and confidential. Candidates are encouraged to utilize this resource and to discuss and clarify questions regarding the areas of the Body of Knowledge where they lack familiarity or seek better understanding.

CMT Level 1 Sample Questions – Candidate Answer Sheet

1. _____	31. _____	61. _____	91. _____
2. _____	32. _____	62. _____	92. _____
3. _____	33. _____	63. _____	93. _____
4. _____	34. _____	64. _____	94. _____
5. _____	35. _____	65. _____	95. _____
6. _____	36. _____	66. _____	96. _____
7. _____	37. _____	67. _____	97. _____
8. _____	38. _____	68. _____	98. _____
9. _____	39. _____	69. _____	99. _____
10. _____	40. _____	70. _____	100. _____
11. _____	41. _____	71. _____	101. _____
12. _____	42. _____	72. _____	102. _____
13. _____	43. _____	73. _____	103. _____
14. _____	44. _____	74. _____	104. _____
15. _____	45. _____	75. _____	105. _____
16. _____	46. _____	76. _____	106. _____
17. _____	47. _____	77. _____	107. _____
18. _____	48. _____	78. _____	108. _____
19. _____	49. _____	79. _____	109. _____
20. _____	50. _____	80. _____	110. _____
21. _____	51. _____	81. _____	111. _____
22. _____	52. _____	82. _____	112. _____
23. _____	53. _____	83. _____	113. _____
24. _____	54. _____	84. _____	114. _____
25. _____	55. _____	85. _____	115. _____
26. _____	56. _____	86. _____	116. _____
27. _____	57. _____	87. _____	117. _____
28. _____	58. _____	88. _____	118. _____
29. _____	59. _____	89. _____	119. _____
30. _____	60. _____	90. _____	120. _____

1. Annie Jones is the portfolio manager for a large-cap equity investment fund. Her colleagues at another investment fund share details about a small-cap company that is now showing a long-term basing pattern on the price charts, and whose prospects for future growth and earnings are outstanding. She invests in the company on behalf of the fund, and over the course of the next three years, the performance of the company's stock contributes greatly to her investment fund beating its benchmark. Jones:
 - A. has violated the CFA Institute Code and Standards because her investment in the small-cap company is contrary to the stated objectives of the fund.
 - B. has not violated the CFA Institute Code and Standards because she conducted additional research based on the information and found supporting evidence of her trade action.
 - C. has violated the CFA Institute Code and Standards because she was made aware of the opportunity by her colleagues at another firm, thereby using confidential information in the process.
 - D. has not violated the CFA Institute Code and Standards because the investment was in the common stock of a listed entity, which is allowed by the investment fund she manages.
2. Which of the following actions is most likely to comply with the CFA Institute Code and Standards?
 - A. Posting information about questions from a prior year's exam on a public discussion forum
 - B. Join discussion groups on LinkedIn/Facebook to discuss and share exam questions
 - C. Posting derogatory comments regarding the CMT® Program on a website for CMT candidates
 - D. Providing your exam prep course provider with a list of topics not tested from the body of knowledge
3. Maggie manages several client portfolios. She utilizes the research of highly skilled analysts in her firm to support her investment decisions. Maggie:
 - A. complied with the CFA Institute Code and Standards.
 - B. violated the CFA Institute Code and Standards because she failed to disclose this information to her clients.
 - C. violated the CFA Institute Code and Standards because she did not exercise due diligence in making her investment decisions.
 - D. violated the CFA Institute Code and Standards because she was unfamiliar with the processes and procedures used by the analysts to arrive at recommendations.

4. Which of the following is most likely to be considered a “material” piece of information regarding a company?
- a) A former board member predicting a long-term decline in the company’s stock
 - b) The loss of a customer representing a significant portion of a company’s gross sales
 - c) The loss of a customer representing a negligible portion of a company’s gross sales
 - d) A competitor CEO’s speculation of a pending management change at the company
5. Qi Lee received a year-end gift of an expensive Rolex watch from a client who was delighted by the performance of his portfolio, which was managed by Lee. Lee should:
- a) decline the gift, as it is prohibited under the CFAI Code of Ethics and Standards.
 - b) decline the gift, as it may have an unintended impact on future performance.
 - c) accept the gift, keeping the matter confidential.
 - d) accept the gift, and disclose it to his employer.
6. When the New High-New Low Index is above its centerline, it shows that:
- a) it is better to step aside and wait for another trade.
 - b) the Market may consolidate near-term, but eventually rise.
 - c) market leaders are bullish and it is better to trade from the long side.
 - d) market leaders are bullish, but it is time to take profits on long positions.
7. If the New High-New Low Index declines while the broad market stays flat or rallies, it is time to:
- a) initiate fresh longs.
 - b) initiate fresh shorts.
 - c) take profits on short positions.
 - d) take profits on long positions.
8. The Advance/Decline line is calculated by:
- a) subtracting the number of declining stocks from advancing stocks and ignoring unchanged stocks.
 - b) subtracting the number of advancing stocks from declining stocks and ignoring unchanged stocks.
 - c) subtracting the number of advancing stocks and unchanged stocks from declining stocks.
 - d) subtracting the number of advancing stocks from declining stocks and adding back the unchanged stocks.

9. If the stock market reaches a new peak while the A/D line does not, it indicates that:
- a) the market may still continue to move higher.
 - b) many stocks are participating and the rally may still continue.
 - c) many stocks are participating and the rally may be near its end.
 - d) fewer stocks are participating and the rally may be near its end.
10. When analyzing candlestick price charts, the appearance of a long body in the same direction as the existing trend indicates:
- a) a potential reversal.
 - b) rising momentum.
 - c) continuation.
 - d) consolidation.
11. Identify the candlestick pattern in which the opening and closing prices are very close or same.
- a) Doji
 - b) Marubozu
 - c) Piercing line
 - d) Dark cloud cover
12. When a series of short candlesticks appears after a period of relatively long sessions, it implies a:
- a) reversal.
 - b) sell signal.
 - c) rising momentum.
 - d) weakening momentum.
13. A candlestick that has a long upper shadow, emerging after an extended uptrend, suggests that:
- a) sellers have lost momentum.
 - b) buyers have lost momentum.
 - c) both buyers and sellers have lose momentum.
 - d) lack of decisive control by either buyers or sellers.
14. Momentum measures:
- a) the rate of change in price.
 - b) the change in price.
 - c) investor psychology.
 - d) investor Sentiment.

15. Lower-priced securities tend to have _____ ATR (Average True Range) and higher-priced securities tend to have _____ ATR.

- a) higher, higher
- b) lower, higher
- c) higher, lower
- d) lower, lower

16. Average True Range (ATR) is an indicator that measures:

- a) volume.
- b) breadth.
- c) volatility.
- d) momentum.

17. A breakaway gap usually:

- a) provides a major divergence signal.
- b) signals the beginning of a new trend.
- c) occurs at the end of an important price move.
- d) occurs during the accumulation phase of the market cycle.

18. Common gaps usually appear in a:

- a) trading range.
- b) bullish market.
- c) bearish market.
- d) trending market.

19. During a consolidation phase, price action normally:

- a) attempts a breakout within three weeks.
- b) attempts a breakout within three months.
- c) moves within well-defined neutral boundaries on low volume.
- d) moves within well-defined neutral boundaries on high volume.

20. An exponential moving average:

- a) gives more weight to the most recent observation.
- b) gives less weight to the most recent observation.
- c) gives equal weight to all observations.
- d) does not suffer from any lag.

21. Gaps normally occur:

- a) between the close of the market on one day and the open of the next day.
- b) between the close of the market on one day and the close of the next day.
- c) between the open of the market on one day and the open of the next day.
- d) between the high of the market on one day and the open of the next day.

22. Which of the following measures makes use of price and volume in its calculation?

- a) Plurality Index
- b) The Arms Index
- c) Relative Strength Index (RSI)
- d) The Money Flow Index (MFI)

23. Which of the following types of moving averages drops off an old data point when a new one is added?

- a) Simple Moving Average
- b) Weighted Moving Average
- c) Triangular Moving Average
- d) Exponential Moving Average

24. MACD line is calculated by:

- a) subtracting the 26-day simple moving average from the 12-day simple moving average.
- b) subtracting the 12-day simple moving average from the 26-day simple moving average.
- c) subtracting the 26-day exponential moving average from the 12-day exponential moving average.
- d) subtracting the 12-day exponential moving average from the 26-day exponential moving average.

25. The basis of OBV is the belief that:

- a) price precedes volume.
- b) volume precedes price.
- c) volume matters most during breakouts.
- d) volume and price are typically coincident indicators.

26. Top-down analysis refers to an analysis of:

- a) monetary policy.
- b) company fundamentals.
- c) the overall economic and business condition.
- d) a specific government, company, (or) asset.

27. Government and companies raise money from investors in:

- a) money markets.
- b) primary markets.
- c) secondary markets.
- d) both primary and secondary markets.

28. Calculate the nominal return based on the information provided below:

- The S&P 500 index fell 0.003 percent in 2015.
- Dividends paid during the year equaled 2.089 percent of the index's value.
- Inflation was 3.4 percent, based on the change in the Consumer Price Index (CPI) for the 10 months ended in November.

- a) -2.08%
- b) -1.3%
- c) 1.30%
- d) 2.08%

29. _____ affects the amount of funds available to the economy as well as their cost.

- a) Credit policy
- b) Economic policy
- c) Monetary policy
- d) Investment policy

30. Yield refers to the projected annual return on a bond based on:

- a) The interest rate of the bond.
- b) the current price of the bond.
- c) future interest payments.
- d) A and B.

31. When a bond's price is greater than 100, the yield is:

- a) lower than the prime rate.
- b) lower than the coupon rate.
- c) higher than the coupon rate.
- d) lower than the discount rate.

32. Treasury inflation-protected securities (TIPS) are bonds whose _____ is adjusted as the consumer price index fluctuates:

- a) principal
- b) interest rate
- c) future value
- d) discount rate

33. An inverted yield curve has historically pointed toward:

- a) economic expansions.
- b) inflationary pressures.
- c) deflationary pressures.
- d) slower economic growth and recessions.

34. Treasury bill returns come from:

- a) volatility.
- b) price changes.
- c) interest payments.
- d) dividend payments

35. The treasury yield curve is usually:

- a) flat.
- b) humped.
- c) inverted.
- d) upward sloping.

36. _____ are bonds without any collateral backing them other than a company's full faith and credit.

- a) Debentures
- b) Treasury bill
- c) Commercial paper (CP)
- d) Certificates of deposit (CDs)

37. _____ give companies the option to buy them back before maturity.

- a) Treasury bills
- b) Puttable bonds
- c) Callable bonds
- d) Convertible bonds

38. _____ provide investors with the option to swap debt for equity.
- a) Treasury bills
 - b) Putable bonds
 - c) Callable bonds
 - d) Convertible bonds
39. At least _____ touches of each trend line are required for a valid rectangle.
- a) one
 - b) two
 - c) three
 - d) four
40. A symmetrical triangle should be longer than:
- a) one week.
 - b) two weeks.
 - c) three weeks.
 - d) three months.
41. _____ occurs after an upward breakout when prices return to the breakout point or chart pattern boundary within 30 days.
- a) A pullback
 - b) A relief rally
 - c) A throwback
 - d) A dead cat bounce
42. When analyzing long term price movements, it could be helpful to use a _____ chart.
- a) Logarithmic
 - b) Candlestick
 - c) EquiVolume
 - d) Pont and Figure
43. An uptrend line has:
- a) a positive slope and is formed by connecting two or more low points.
 - b) a positive slope and is formed by connecting two or more high points.
 - c) a negative slope and is formed by connecting two or more high points.
 - d) a negative slope and is formed by connecting two or more low points.

44. The key variable in the CAPM (Capital Asset Pricing Model) equation is:

- a) Rho.
- b) Beta.
- c) Delta.
- d) Alpha.

45. A complete Elliott wave cycle is made up of _____ waves.

- a) three
- b) five
- c) eight
- d) thirteen

46. In Elliott wave theory, sub-waves 2 and 4 are called:

- a) Zigzags.
- b) Motive waves.
- c) Impulse waves.
- d) Corrective waves

47. _____ is a sideways, three-wave corrective pattern.

- a) Flat
- b) Zigzag
- c) Triangle
- d) Ending Diagonal

48. Open interest in a futures contract represents:

- a) the number of contracts traded in a day.
- b) the sum of total volume and open interest.
- c) the number of outstanding contracts on any given day.
- d) the number of outstanding contracts in first day of the month.

49. Which of the following is a CORRECT interpretation of open interest?

- a) Rising open interest in an uptrend is bearish.
- b) Rising open interest in a downtrend is bullish.
- c) Declining open interest in an uptrend is bearish.
- d) Declining open interest in a downtrend is bearish.

50. Force Index is calculated by:

- a) subtracting the previous day's close from the current day's close and multiplying by volume.
- b) subtracting the current day's close from the previous day's close and multiplying by volume.
- c) subtracting the previous day's close from the current day's close and multiplying by the 10-day average volume.
- d) subtracting the previous day's close from the current day's close and multiplying by the 10-day average volume.

51. A protective stop is:

- a) an order to sell below the market when you are long.
- b) a stop-loss order that protects you from getting badly hurt by a reversal.
- c) an order to cover the short positions above the market when you are short.
- d) All of the above

52. In constructing a Point & Figure chart, a new box is added only when:

- a) price has moved to a new high.
- b) volume confirms the price movement.
- c) price has moved by less than the specified box size.
- d) price has moved by equal to or more than the specified box size.

53. Violated support levels typically:

- a) lose importance after three weeks.
- b) indicate an imminent price reversal.
- c) lose importance after three months.
- d) become overhead resistance on price bounces.

54. A rising relative strength line for a stock in a falling market indicates that:

- a) price and volume are diverging.
- b) the stock is performing worse than the market.
- c) the stock is performing better than the market (on a relative basis).
- d) the stock is performing better than the market (on an absolute basis).

55. Which of the following lists arranges cycles from shortest to longest in duration?

- a) Presidential, seasonal, Kondratieff
- b) Kondratieff, Presidential, seasonal
- c) Seasonal, Presidential, Kondratieff
- d) Presidential, Kondratieff, seasonal

56. A narrowing of Bollinger Bands normally indicates that:
- a) a stock is ready for a rally.
 - b) a stock is ready for a decline.
 - c) a stock's volatility has increased.
 - d) a stock's volatility has decreased.
57. A company whose stock is in a well-established uptrend, reports earnings slightly above consensus estimates. After an initial rally, the stock then ends lower on the day. This would probably be an example of:
- a) market discounting.
 - b) investor under reaction.
 - c) unrealistic expectations.
 - d) a short-coming of fundamental analysis.
58. To increase the sensitivity of a 5×3 Point & Figure chart, one would:
- a) increase the box size.
 - b) decrease the box size.
 - c) increase the number of boxes needed for a reversal.
 - d) opt for a one-box reversal type Point & Figure chart.
59. You are updating a one box reversal Point & Figure chart. The box size is \$1 and the current column has 4 X's. The most recent posting is \$40. The stock closed yesterday at \$40.50. Today, the stock opened at \$40.60, ran up to \$40.95, and closed at the low for the day at \$39.11. How many new postings will you be making on the chart to reflect today's trading?
- a) 0
 - b) 1
 - c) 2
 - d) 3
60. Cycles are usually measured from:
- a) curl to curl.
 - b) high to high.
 - c) crest to crest.
 - d) trough to trough.

61. In a 5-day simple moving average, each day's price is assigned a weighting of:

- a) 5%.
- b) 10%.
- c) 20%.
- d) 25%.

62. Volume divergences occur when:

- a) volume expands while open interest contracts.
- b) volume declines during a sideways congestion pattern.
- c) volume expands while the daily trading range contracts.
- d) a new high in an uptrend takes place on declining volume.

63. The primary use of a candlestick pattern is to:

- a) determine if you are in a bull market or a bear market.
- b) define the primary, intermediate, and short-term trends.
- c) strengthen the possibility of a reversal or continuation of a price trend.
- d) define the intermediate and short-term trends, but not the primary trend.

64. Which is an example of candle pattern filtering?

- a) Using cycles to confirm the existence of a pattern
- b) Using western techniques to confirm a candle pattern
- c) Accepting only those patterns that fit the strict definition of their pattern
- d) Accepting only those patterns that are confirmed by the intermediate-term trend

65. A Point & Figure chart:

- a) ignores time.
- b) is limited by a four-year time horizon.
- c) cannot be used to project price targets.
- d) gives a clear picture of volume distribution.

66. The Kondratieff cycle extends for a period of about:

- a) 13 years.
- b) 50 years.
- c) 54 years.
- d) 60 years

67. A weak economy is usually favorable for which of the following asset classes?

- a) Bonds
- b) Stocks
- c) Real estate
- d) Commodities

68. Trend lines:

- a) are not useful for short-term trading.
- b) offer late confirmation of a change in trend.
- c) are more significant the longer they stay in force.
- d) can only be determined by least square regression method.

69. Which of the following is a consolidation pattern?

- a) Flag
- b) Diagonal
- c) Rising wedge
- d) Broadening formation

70. Seasonality is a cycle that occurs:

- a) yearly.
- b) weekly.
- c) monthly.
- d) quarterly.

71. The interchangeability of a financial instrument is normally referred to as:

- a) hedging.
- b) fungibility.
- c) arbitraging.
- d) backwardation.

72. The VIX index measures:

- a) the implied volatility of the S&P 100 index options.
- b) the historical volatility of the S&P 500 index options.
- c) the historical volatility of the S&P 100 index options.
- d) the implied volatility of the S&P 500 index options.

73. If the VIX is quoted at 20, it indicates the market is expecting a movement of about _____ percent over the next 30 days.

- a) 3.77
- b) 5.77
- c) 3.33
- d) 5.07

74. In a double bottom pattern, volume is generally:

- a) rising throughout the pattern.
- b) declining throughout the pattern.
- c) lower on the left bottom than the right.
- d) higher on the left bottom than the right.

75. When an economic crisis begins, investors generally buy _____ securities for their relative safety and sell _____ securities.

- a) corporate, treasury
- b) treasury, corporate
- c) commercial papers, corporate
- d) corporate, TIPS (Treasury Inflation-Protected Securities)

76. The _____ index is a popular benchmark for measuring the performance of small-cap companies.

- a) S&P 500
- b) Russell 2000
- c) NASDAQ Composite
- d) Dow Jones Industrial Average

77. The NASDAQ Composite index is a _____ weighted index.

- a) market-cap
- b) price
- c) trade
- d) equal

78. The quadratic mean is often used for an estimation of:

- a) error.
- b) return.
- c) growth.
- d) volatility.

79. The harmonic mean is a _____ weighted average.
- a) price
 - b) time
 - c) trade
 - d) volume
80. Bullish support lines on a 3-box reversal Point & Figure chart are drawn at a _____ degree angle:
- a) 45
 - b) 60
 - c) 75
 - d) 90
81. If the fund has a beta of 1.2, then it has _____ than the overall market.
- a) 2% more volatility
 - b) 20% less volatility
 - c) 0.2% more volatility
 - d) 20% more volatility
82. An effective way of detrending price data to identify cycles is to:
- a) use an oscillator and identify extreme levels.
 - b) use a moving average of the highs/lows over n periods.
 - c) divide the closing prices by a moving average of those prices.
 - d) plot a linear regression (best-fit) line through the data midpoint.
83. An important feature of the Martingale process that makes it an appropriate model for EMH is:
- a) the absence of future price predictability.
 - b) the absence of price return anomalies that are unexplained.
 - c) the presence of price return anomalies that are unexplained.
 - d) the presence of information that aids in future price predictability.
84. In EMH, technical traders can be thought of as a specific type of noise trader because:
- a) technical traders are rational traders.
 - b) technical traders are not rational traders.
 - c) a positive feedback loop can lead to pronounced bubbles.
 - d) the cancel-out phenomenon will apply to a large number of trend following strategies.

85. The stochastics indicator measures:

- a) where today's typical price fits into the recent trading range.
- b) the distance in percentage between the first and last values over n-days.
- c) the relative position of the closing price within a past high-low range.
- d) the relative strength of the current price movement as it increases from 0 to 100.

86. Which of the following statements is an important assumption of the Efficient Market Hypothesis?

- a) Investors as a group will act irrationally.
- b) Investors as a group will act rationally.
- c) Prices typically do not reflect all available information.
- d) Liquidity constraints may sometimes drive prices away from their intrinsic value.

87. Slippage is normally attributed to:

- a) liquidity risk.
- b) default risk.
- c) market risk.
- d) credit risk.

88. Objective technical analysis methods:

- a) normally witness less drawdown.
- b) normally witness high drawdown.
- c) require a disciplined approach for success.
- d) are well-defined procedures that issue unambiguous signals.

89. When a great majority of investors have the same view of the market:

- a) market sentiment is positive.
- b) it is safe to assume that the market may trend to further highs.
- c) it is a warning to expect a reversal contrary to the majority opinion.
- d) it is safe to assume that a reversal of trend is a low-probability outcome.

90. The Fed largely controls:

- a) money supply.
- b) long-term interest rates.
- c) short-term interest rates.
- d) both money supply and long-term interest rates.

91. Which of the following scenarios is generally followed by weakness in equity markets?

- a) Higher interest rates
- b) Lower interest rates
- c) Slower money-supply growth
- d) Both higher interest rates and slower money-supply growth

92. Monetary indicators are considered _____ indicators.

- a) internal
- b) external
- c) sentiment
- d) momentum

93. The concept of randomness in price action is directly related to the:

- a) Dow Theory
- b) Elliott wave theory
- c) Pythagorean theorem
- d) evolution of the probability theory

94. As per the theory of contrary opinion, a high degree of consensus normally precedes a:

- a) trend reversal
- b) trend continuation
- c) short-term consolidation
- d) pause within an existing trend

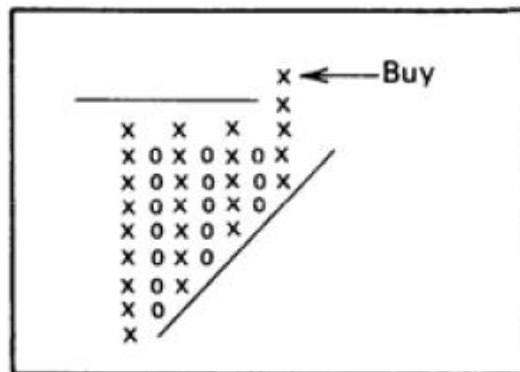
95. The white or black portion of a candlestick is also referred to as a/the:

- a) Doji
- b) Shadow
- c) Real body
- d) Spinning top

96. _____ has the highest component weighting in the U.S. Dollar index.

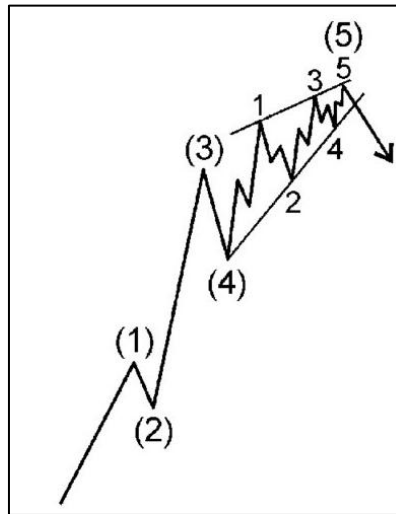
- a) Euro
- b) Japanese Yen
- c) British pound
- d) Canadian dollar

97. The square root of variance is also referred to as:
- Mean deviation.
 - Quadratic mean.
 - Harmonic mean.
 - Standard deviation.
98. When a larger number of stocks register new lows vs. new highs, the New High-New Low Index is:
- Positive, and is below the centerline.
 - Positive, and is above the centerline.
 - Negative, and is above the centerline.
 - Negative, and is below the centerline.
99. The “*Percentage of stocks above their 50-day MA*” indicator is considered to be overbought, when it crosses above:
- 50%.
 - 80%.
 - 75%.
 - 60%.
100. Identify the Point & Figure pattern highlighted below:



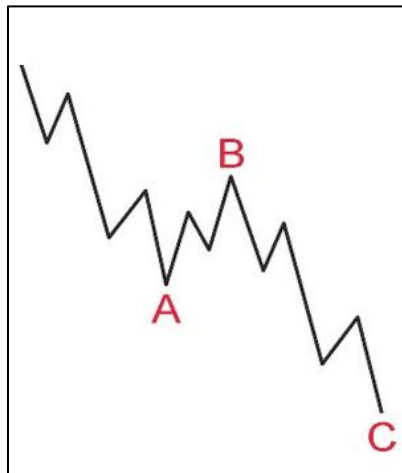
- Quadruple Top Buy
- Upside Breakout of a Descending Triangle
- Upside Breakout of a Symmetrical Triangle
- Upside Breakout of an Ascending Triangle

101. The Elliott Wave pattern highlighted below is a/an:



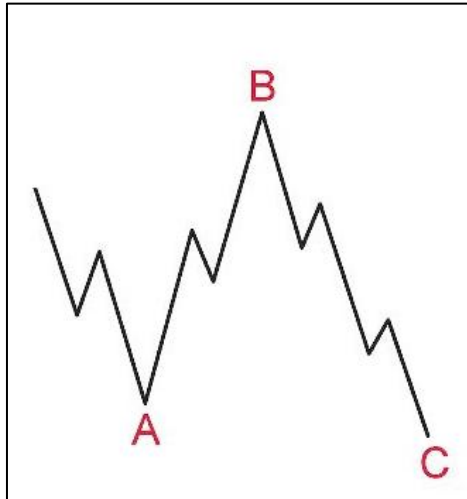
- a) Triangle
- b) Rising wedge
- c) Ending Diagonal
- d) Leading Diagonal

102. Identify the Elliott Wave pattern highlighted below:



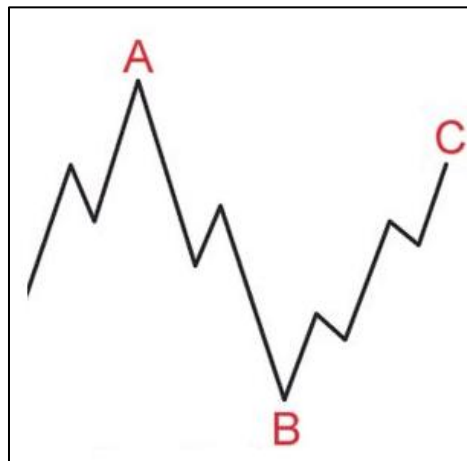
- a) Flat in a bull market
- b) Flat in a bear market
- c) Zigzag in a bull market
- d) Zigzag in a bear market

103. The Elliott Wave pattern highlighted below is a/an:



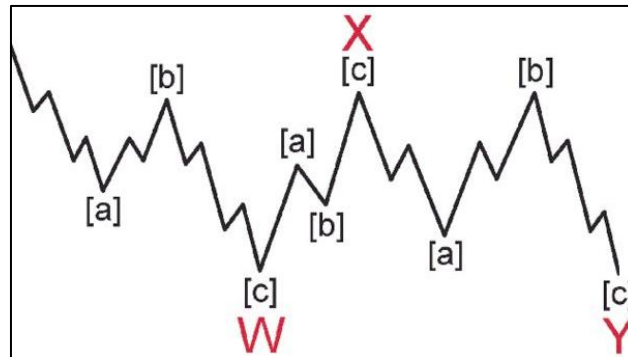
- a) Flat
- b) Regular Flat
- c) Running Flat
- d) Expanded Flat

104. Identify the Elliott Wave pattern highlighted below:



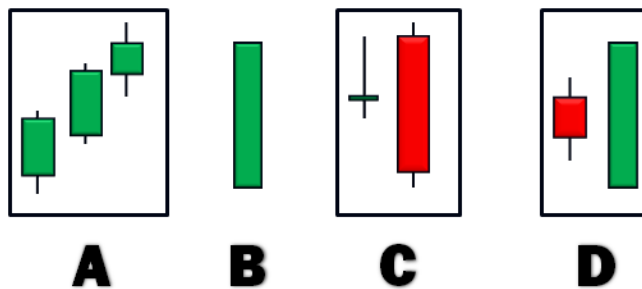
- a) Expanded Flat
- b) Running Flat
- c) Regular Flat
- d) Zigzag

105. Identify the following “Double-Three” combination pattern as per the Elliott Wave Theory:



- a) Zigzag, Flat, Flat
- b) Zigzag, Zigzag, Flat
- c) Zigzag, Flat, Zigzag
- d) Flat, Zigzag, Zigzag

106. Identify the CORRECT white marubozu candlestick pattern:



- a) A
- b) B
- c) C
- d) D

107. The chart formation highlighted below is:



- a) An island bottom.
- b) An ascending triangle.
- c) A descending triangle.
- d) A symmetrical triangle.

108. The chart formation highlighted below is:



- a) A triple bottom.
- b) A double bottom.
- c) An island bottom.
- d) A three Buddha pattern.

109. Identify the candlestick pattern highlighted (inside the blue circle) in the chart:



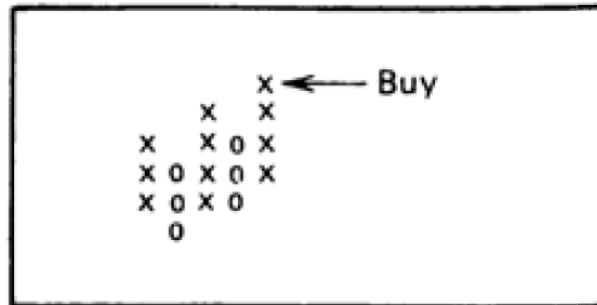
- a) Hammer
- b) Evening star
- c) Hanging man
- d) Inverted hammer

110. The chart formation highlighted below is a:



- a) Hammer
- b) Morning star
- c) Hanging man
- d) Bullish engulfing

111. Identify the Point & Figure pattern highlighted below:



- a) Triple Top Buy
- b) Bullish Catapult
- c) Ascending Triple Top
- d) Bullish Ascending Triangle Breakout

112. Identify the gap highlighted (inside the blue circle) in the chart:



- a) Common gap
- b) Runaway gap
- c) Breakaway gap
- d) Exhaustion gap

113. The chart formation highlighted below is:



- a) A Triple bottom.
- b) An Island bottom.
- c) A Head and Shoulders pattern.
- d) An Inverted Head and Shoulders pattern.

114. Identify the gap highlighted (black arrow) in the chart:



- a) Common gap
- b) Runaway gap
- c) Exhaustion gap
- d) Breakaway gap

115. Identify the chart formation highlighted below:



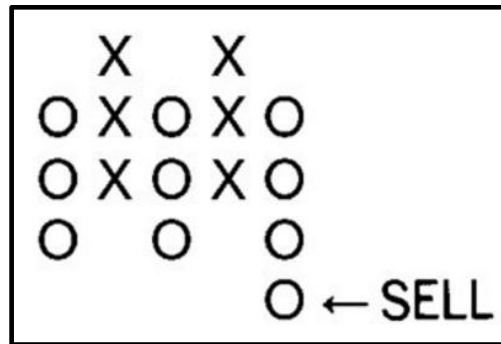
- a) Blow off
- b) Bull Trap
- c) Bear Trap
- d) Bullish Breakout

116. The chart formation highlighted below is a:



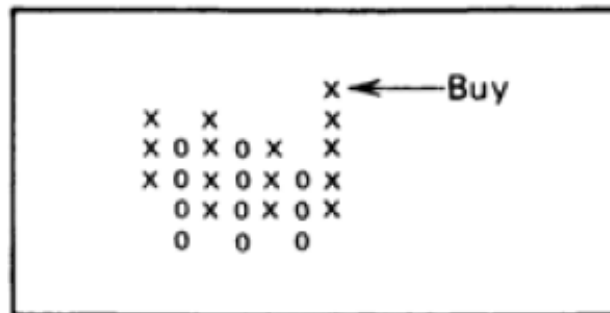
- a) Flag.
- b) Pennant.
- c) Triangle.
- d) Rising wedge.

117. Identify the Point & Figure pattern highlighted below:



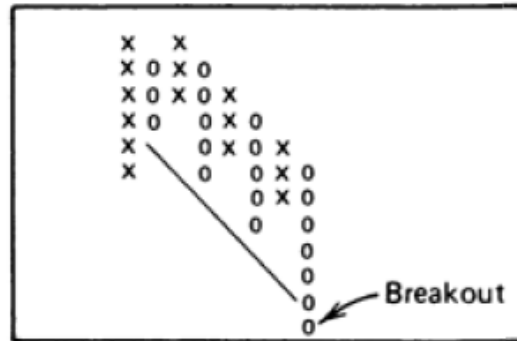
- a) Triple bottom
- b) Symmetric triangle
- c) Descending triangle
- d) Descending triple bottom

118. The Point & Figure pattern highlighted below is a:



- a) Triple Top Buy
- b) Bullish Catapult Buy
- c) Spread Triple Top Buy
- d) Ascending Triple Top Buy

119. The Point & Figure pattern highlighted below is a:



- a) Bearish Catapult.
- b) Double Bottom Sell.
- c) Downside Breakout of a Bullish Support Line.
- d) Downside Breakout of a Bearish Support Line.

120. Identify the chart formation highlighted below:



- a) Triple Top
- b) Rising Wedge
- c) Rounding Top
- d) Head and Shoulders top

**CMT Level I
Sample Exam
Answers**

1. Annie Jones is the portfolio manager for a large-cap equity investment fund. Her colleagues at another investment fund share details about a small-cap company that is now showing a long-term basing pattern on the price charts, and whose prospects for future growth and earnings are outstanding. She invests in the company on behalf of the fund, and over the course of the next three years, the performance of the company's stock contributes greatly to her investment fund beating its benchmark. Jones
 - a. has violated the CFA Institute Code and Standards because her investment in the small-cap company is contrary to the stated objectives of the fund.Reference: CFA Institute Code of Ethics and Standards
2. Which of the following actions is most likely to comply with the CFA Institute Code and Standards?
 - d. Providing your exam prep course provider with a list of topics not tested from the body of knowledgeReference: CFA Institute Code of Ethics and Standards
3. Maggie manages several client portfolios. She utilizes the research of highly skilled analysts in her firm to support her investment decisions. Maggie:
 - a. complied with the CFA Institute Code and Standards.Reference: CFA Institute Code of Ethics and Standards
4. Which of the following is most likely to be considered a "material" piece of information regarding a company?
 - b. The loss of a customer representing a significant portion of a company's gross sales.Reference: CFA Institute Code of Ethics and Standards
5. Qi Lee received a year-end gift of an expensive Rolex watch from a client who was delighted by the performance of his portfolio, which was managed by Lee. Lee should:
 - d. accept the gift, and disclose it to his employer.Reference: CFA Institute Code of Ethics and Standards
6. When the New High-New Low Index is above its centerline, it shows that:
 - c. market leaders are bullish and it is better to trade from the long side.Reference: MTA, CMT Level I Curriculum (2016), Chapter 6
7. If the New High-New Low Index declines while the broad market stays flat or rallies, it is time:
 - d. to take profits on long positions.Reference: MTA, CMT Level I Curriculum (2016), Chapter 6
8. The Advance/Decline line is calculated by:
 - a. subtracting the number of declining stocks from advancing stocks and ignoring unchanged stocks.Reference: MTA, CMT Level I Curriculum (2016), Chapter 8

9. If the stock market reaches a new peak while the A/D line does not, it indicates that:
d. fewer stocks are participating and the rally may be near its end.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 8
10. When analyzing candlestick price charts, the appearance of a long body in the same direction as the existing trend indicates:
c. continuation.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
11. Identify the candlestick pattern in which the opening and closing prices are very close or same:
a. Doji
Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
12. When a series of short candlesticks appears after a period of relatively long sessions, it implies a:
d. weakening momentum.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
13. A candlestick that has a long upper shadow, emerging after an extended uptrend, suggests that:
b. Buyers have lost momentum.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
14. Momentum measures:
a. the rate of change in price.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11
15. Lower-priced securities tend to have _____ ATR (Average True Range) and higher-priced securities tend to have _____ ATR.
b. lower, higher
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11
16. Average True Range (ATR) is an indicator that measures:
c. volatility.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11
17. A breakaway gap usually:
b. signals the beginning of a new trend.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11
18. Common gaps usually appear in a:
a. trading range.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11

19. During a consolidation phase, price action normally:
c. moves within well-defined neutral boundaries on low volume.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11
20. An exponential moving average:
a. gives more weight to the most recent observation.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 12
21. Gaps normally occur:
a. Between the close of the market on one day and the open of the next day
Reference: MTA, CMT Level I Curriculum (2016), Chapter 12
22. Which of the following measures make use of price and volume in its calculation?
d. The Money Flow Index (MFI)
Reference: MTA, CMT Level I Curriculum (2016), Chapter 13
23. Which of the following types of moving averages drops off an old data point when a new one is added?
a. Simple moving average
Reference: MTA, CMT Level I Curriculum (2016), Chapter 13
24. MACD line is calculated by:
c. subtracting the 26-day exponential moving average from the 12-day exponential moving average.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 13
25. The basis of OBV is the belief that:
b. volume precedes price.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 13
26. Top-down analysis refers to an analysis of:
c. the overall economic and business condition.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 16
27. Government and companies raise money from investors in:
b. primary markets.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 16

28. An effective way of detrending price data to identify cycles is to: Calculate the nominal return based on the information provided below:
- The S&P 500 index fell 0.003 percent in 2015.
 - Dividends paid during the year equaled 2.089 percent of the index's value.
 - Inflation was 3.4 percent, based on the change in the Consumer Price Index (CPI) for the 10 months ended in November.
- d. 2.08%
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 16
29. _____ affects the amount of funds available to the economy as well as their cost.
- c. Monetary policy
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
30. Yield refers to the projected annual return on a bond based on:
- d. A and B.
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
31. When a bond's price is greater than 100, the yield is:
- b. lower than the coupon rate.
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
32. Treasury inflation-protected securities (TIPS) are bonds whose _____ is adjusted as the consumer price index fluctuates:
- a. principal
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
33. An inverted yield curve has historically pointed toward:
- d. slower economic growth and recessions.
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
34. Treasury bill returns come from:
- b. price changes.
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
35. The treasury yield curve is usually:
- d. upward sloping.
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
36. _____ are bonds without any collateral backing them other than a company's full faith and credit.
- a. Debentures
- Reference: MTA, CMT Level I Curriculum (2016), Chapter 18

37. _____ give companies the option to buy them back before maturity.
c. Callable Bonds
Reference: MTA, CMT Level I Curriculum (2016), Chapter 18
38. _____ provide investors with the option to swap debt for equity.
d. Convertible bonds
Reference: MTA, CMT Level I Curriculum (2016), Chapter 18
39. At least _____ touch(es) of each trend line are required for a valid rectangle.
b. two
Reference: MTA, CMT Level I Curriculum (2016), Chapter 48
40. A symmetrical triangle should be longer than:
c. three weeks.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 48
41. _____ occurs after an upward breakout when prices return to the breakout point or chart pattern boundary within 30 days.
c. A throwback
Reference: MTA, CMT Level I Curriculum (2016), Chapter 47
42. When analyzing long term price movements, it could be helpful to use a _____ chart.
a. logarithmic
Reference: MTA, CMT Level I Curriculum (2016), Chapter 46
43. An uptrend line has:
a. a positive slope and is formed by connecting two or more low points.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 46
44. The key variable in the CAPM (Capital Asset Pricing Model) equation is:
b. Beta.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 42
45. A complete Elliott wave cycle is made up of _____ waves.
c. eight
Reference: MTA, CMT Level I Curriculum (2016), Chapter 39
46. In Elliott wave theory, sub-waves 2 and 4 are called:
d. Corrective Waves.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 39
47. _____ is a sideways, three-wave corrective pattern.
a. Flat
Reference: MTA, CMT Level I Curriculum (2016), Chapter 39

48. Open interest represents:
c. the number of outstanding contracts on any given day.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 38
49. Which of the following is a CORRECT interpretation of open interest?
c. Declining open interest in an uptrend is bearish.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 38
50. Force Index is calculated by:
a. subtracting the previous day's close from the current day's close and multiplying by volume.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 37
51. A protective stop is:
d. All of the above.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 34
52. In constructing a Point & Figure chart, a new box is added only when:
d. price has moved by equal to or more than the specified box size.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 22
53. Violated support levels typically:
d. become overhead resistance on price bounces.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 34
54. A rising relative strength line for a stock in a falling market indicates that:
c. the stock is performing better than the market (on a relative basis).
Reference: MTA, CMT Level I Curriculum (2016), Chapter 54
55. Which of the following lists arranges cycles from shortest to longest in duration?
c. Seasonal, Presidential, Kondratieff
Reference: MTA, CMT Level I Curriculum (2016), Chapter 53
56. A narrowing of Bollinger Bands normally indicates that:
d. a stock's volatility has decreased.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 11
57. A company whose stock is in a well-established uptrend, reports earnings slightly above consensus estimates. After an initial rally, the stock then ends lower on the day. This would probably be an example of:
a. Market discounting.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 2

58. To increase the sensitivity of a 5×3 Point & Figure chart, one would:

b. decrease the box size.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 22

59. You are updating one box reversal Point & Figure chart. The box size is \$1 and the current column has 4 X's. The most recent posting is \$40. The stock closed yesterday at \$40.50. Today, the stock opened at \$40.60, ran up to \$40.95, and closed at the low for the day at \$39.11. How many new postings will you be making on the chart to reflect today's trading?

a. 0 (zero)

Reference: MTA, CMT Level I Curriculum (2016), Chapter 22

60. Cycles are usually measured from:

d. trough to trough.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 53

61. In a 5-day simple moving average, each day's price is assigned a weighting of:

c. 20%.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 13

62. Volume divergences occur when:

d. a new high in an uptrend takes place on declining volume.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 13

63. The primary use of a candlestick pattern is to:

c. strengthen the possibility of a reversal (or) continuation of a price trend.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 10

64. Which is an example of candle pattern filtering?

b. Using western techniques to confirm a candle pattern

Reference: MTA, CMT Level I Curriculum (2016), Chapter 10

65. A Point & Figure chart:

a. ignores time.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 22

66. The Kondratieff cycle extends for a period of about:

c. 54 years.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 53

67. A weak economy is usually favorable for which of the following asset classes?

a. Bonds

Reference: MTA, CMT Level I Curriculum (2016), Chapter 17

68. Trend lines:
c. are more significant the longer they stay in force.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 46
69. Which of the following is a consolidation pattern?
a. Flag
Reference: MTA, CMT Level I Curriculum (2016), Chapter 46
70. Seasonality is a cycle that occurs:
a. yearly.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 53
71. The interchangeability of a financial instrument is normally referred to as:
b. fungibility.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 44
72. The VIX index measures:
d. the implied volatility of the S&P 500 index options.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 52
73. If the VIX is quoted at 20, it indicates the market expecting a movement of about _____ percent over the next 30 days.
b. 5.77
Reference: MTA, CMT Level I Curriculum (2016), Chapter 52
74. In a double bottom pattern, volume is generally _____.
d. higher on the left bottom than the right.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 12
75. When an economic crisis begins, investors generally buy _____ securities for their relative safety and sell _____ securities.
b. treasury, corporate
Reference: MTA, CMT Level I Curriculum (2016), Chapter 17
76. The _____ index is a popular benchmark for measuring the performance of small-cap companies.
b. Russell 2000
Reference: MTA, CMT Level I Curriculum (2016), Chapter 20
77. The NASDAQ Composite index is a _____ weighted index.
a. Market-cap
Reference: MTA, CMT Level I Curriculum (2016), Chapter 20

78. The quadratic mean is often used for an estimation of:
a. error.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 21
79. The harmonic mean is a _____ weighted average.
b. time
Reference: MTA, CMT Level I Curriculum (2016), Chapter 21
80. Bullish support lines on a 3-box reversal Point & Figure chart are drawn at a _____ degree angle:
a. 45
Reference: MTA, CMT Level I Curriculum (2016), Chapter 22
81. If a fund has a beta of 1.2, then it has _____ than the overall market.
d. 20% more volatility
Reference: MTA, CMT Level I Curriculum (2016), Chapter 21
82. An effective way of detrending price data to identify cycles is to:
c. divide the closing prices by a moving average of those prices.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 21
83. An important feature of the martingale process that makes it an appropriate model for EMH is:
a. the absence of future price predictability.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 41
84. In EMH, technical traders can be thought of as a specific type of noise trader because:
b. technical traders are not rational traders.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 41
85. The stochastics indicator measures:
c. the relative position of the closing price within a past high-low range
Reference: MTA, CMT Level I Curriculum (2016), Chapter 15
86. Which of the following statements is an important assumption of the Efficient Market Hypothesis (EMH)?
b. Investors as a group will act rationally
Reference: MTA, CMT Level I Curriculum (2016), Chapter 42
87. Slippage is normally attributed to:
a. Liquidity risk
Reference: MTA, CMT Level I Curriculum (2016), Chapter 17

88. Objective technical analysis methods:
d. Are well-defined procedures that issue unambiguous signals
Reference: MTA, CMT Level I Curriculum (2016), Chapter 23
89. When a great majority of investors have the same view of the market:
c. It is a warning to expect a reversal contrary to the majority opinion
Reference: MTA, CMT Level I Curriculum (2016), Chapter 26
90. The Fed largely controls:
d. Both money supply and long-term interest rates.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 26
91. Which of the following scenarios are generally followed by weakness in equity markets?
d. Both higher interest rates and slower money-supply growth
Reference: MTA, CMT Level I Curriculum (2016), Chapter 26
92. Monetary indicators are considered _____ indicators.
b. external
Reference: MTA, CMT Level I Curriculum (2016), Chapter 26
93. The concept of randomness in price action is directly related to the:
d. evolution of the probability theory.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 26
94. As per the theory of contrary opinion, a high degree of consensus normally precedes a:
a. trend reversal.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 9
95. The white or black portion of a candlestick is also referred to as a/the:
c. Real body.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
96. _____ has the highest component weighting in the U.S. Dollar index.
a. Euro
Reference: MTA, CMT Level I Curriculum (2016), Chapter 20
97. The square root of variance is also referred to as:
d. standard deviation.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 21
98. When a larger number of stocks register new lows vs. new highs, the New High-New Low Index is:
d. negative, and is below the centerline.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 6

99. The “*Percentage of stocks above their 50-day MA*” indicator is considered to be overbought, when it crosses above:

c. 75%.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 7

100. Identify the Point & Figure pattern highlighted below:

d. Upside Breakout of an Ascending triangle

Reference: MTA, CMT Level I Curriculum (2016), Chapter 22

101. The Elliott Wave pattern highlighted below is a/an:

c. ending Diagonal.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 39

102. Identify the Elliott Wave pattern highlighted below:

c. Zigzag in a bull market

Reference: MTA, CMT Level I Curriculum (2016), Chapter 39

103. The Elliott Wave pattern highlighted below is a/an:

d. expanded Flat.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 39

104. Identify the Elliott Wave pattern highlighted below:

b. Running flat

Reference: MTA, CMT Level I Curriculum (2016), Chapter 39

105. Identify the following “Double-Three” combination pattern as per the Elliott Wave Theory:

b. Zigzag, Zigzag, Flat

Reference: MTA, CMT Level I Curriculum (2016), Chapter 39

106. Identify the CORRECT white marubozu candlestick pattern:

d. B

Reference: MTA, CMT Level I Curriculum (2016), Chapter 10

107. The chart formation highlighted below is:

b. an ascending triangle.

Reference: MTA, CMT Level I Curriculum (2016), Chapter 16

108. The chart formation highlighted below is:

b. A double bottom

Reference: MTA, CMT Level I Curriculum (2016), Chapter 16

109. Identify the candlestick pattern highlighted (inside the blue circle) in the chart:

a. Hammer

- Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
110. The chart formation highlighted below is a:
d. bullish engulfing.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 10
111. Identify the Point & Figure pattern highlighted below:
c. Ascending Triple Top
Reference: MTA, CMT Level I Curriculum (2016), Chapter 22
112. Identify the gap highlighted (inside the blue circle) in the chart:
d. Exhaustion gap
Reference: MTA, CMT Level I Curriculum (2016), Chapter 12
113. The chart formation highlighted below is:
d. an inverted Head & Shoulders pattern.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 50
114. Identify the gap highlighted (black arrow) in the chart:
d. Breakaway gap
Reference: MTA, CMT Level I Curriculum (2016), Chapter 12
115. Identify the chart formation highlighted below:
b. bull trap.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 12
116. The chart formation highlighted below is a:
a. Flag.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 48
117. Identify the Point & Figure pattern highlighted below:
a. Triple bottom
Reference: MTA, CMT Level I Curriculum (2016), Chapter 22
118. The Point & Figure pattern highlighted below is a:
c. Spread Triple Top Buy.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 22
119. The Point & Figure pattern highlighted below is a:
d. Downside breakout of a Bearish Support Line.
Reference: MTA, CMT Level I Curriculum (2016), Chapter 22
120. Identify the chart formation highlighted below:
d. Head & Shoulders top
Reference: MTA, CMT Level I Curriculum (2016), Chapter 50